Ujjwal Patel

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WORK EXPERIENCE

Meta Platforms Inc. **TX, USA**

ML Engineer *Jun 2025 – Aug 2025*

- Developed and deployed AI-driven predictive maintenance models using Python, TensorFlow, and AWS SageMaker, improving equipment uptime by 21%.
- Implemented real-time ML pipelines for sensor data analysis, reducing inference latency by 30% through model compression and GPU optimization.
- Collaborated with cross-functional software engineering teams to integrate computer vision and anomaly detection models into data center monitoring and automated infrastructure inspection systems.
- Built MLOps workflows with Docker, Kubernetes, and MLflow, cutting model deployment time by 40% while ensuring reproducibility and compliance Meta's production ML standards..
- Spearheaded the development of reinforcement learning–based resource optimization, for large-scale distributed computing workloads, improving GPU and compute resource utilization by **15%** while reducing infrastructure costs.

NVIDIA Corporation **India**

AI Engineer *June 2023 - Jun 2024*

- Designed and fine-tuned large language models (LLMs) for content moderation and personalization, boosting classification precision by 17%.
- Leveraged PyTorch, Hugging Face Transformers, and LangChain to create conversational AI agents that enhanced user engagement by 25%.
- Automated training and evaluation pipelines using Airflow and Ray, improving experiment turnaround time by 35%.
- Partnered with product teams to translate business goals into scalable AI solutions, contributing to Meta’s Responsible AI framework.
- Implemented real-time recommendation system enhancements using graph neural networks (GNNs), leading to a 10% increase in user session duration across pilot markets.
- Collaborated with product, engineering, and data science teams to productionize AI capabilities supporting 100K+ active users while achieving 99.9% service availability

IBM Corporation **India**

ML Engineer *Aug 2021 – May 2023*

- Built supervised and unsupervised ML models for client analytics projects using Python, Scikit-learn, and SQL, improving forecast accuracy by 22%.
- Developed NLP-based document classification systems with BERT and spaCy to automate data labeling, reducing manual processing time by 50%.
- Implemented MLOps pipelines on IBM Cloud using Watson Studio, streamlining model deployment and monitoring.
- Collaborated with data engineers and product managers to optimize AI model performance and explainability, driving successful client adoption.
- Created an AI-driven anomaly detection framework for financial transaction monitoring, reducing false positives by 32% and enhancing compliance analytics for enterprise clients.

EDUCATION

University of Texas at Arlington **TX, USA**

Master’s of Computer Science *Graduation Date: May 2026*

Mumbai University **India**

Bachelor’s of Science, Computer Science *Graduation Date: May 2023*

SKILLS & INTERESTS

Programming Languages	**Python, SQL, R, C++, Java**
Machine Learning & AI	**Supervised/Unsupervised Learning, Deep Learning, Computer Vision, Natural Language Processing , Generative AI, Large Language Models , Reinforcement Learning**
Frameworks & Libraries	**TensorFlow, PyTorch, Keras, Scikit-learn, Hugging Face Transformers, LangChain, OpenCV, XGBoost, spaCy**
MLOps & Deployment	**MLflow, Airflow, Docker, Kubernetes, AWS SageMaker, Azure ML, Databricks, CI/CD Pipelines**
Data Engineering & Visualization	**Pandas, NumPy, Spark, SQL, Tableau, Power BI, Matplotlib, Seaborn**
Soft Skills & Collaboration	**Cross-Functional Communication, Problem Solving, Agile Methodologies, Leadership, Technical Documentation**